

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
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Date:	24/08/2026	Duration:	60 minutes

Code of Conduct

I Tamil Elakiya S (192524443) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Tamil Elakiya S

Rubrics for Calculation

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25

Digital Farmer Marketplace: A Digital Platform for Direct Farmer–Buyer Connectivity

1. INTRODUCTION

Agriculture is one of the major sectors supporting the Indian economy and providing livelihood opportunities to a large population. However, farmers frequently face challenges in marketing their agricultural products. Traditional agricultural marketing systems commonly involve multiple intermediaries between farmers and final consumers. This can reduce the farmer's share of the final selling price and create difficulties in obtaining accurate information about market demand and prevailing prices. The growth of information and communication technologies has created opportunities to improve agricultural marketing through digital platforms. E-commerce, mobile applications, online marketplaces, digital payment systems, and real-time information services can connect farmers with buyers more directly. Such systems can improve market accessibility, product visibility, price transparency, and communication between participants.

Research conducted in India has already demonstrated the potential of digital technologies in agricultural marketing. Studies have examined farmer adoption of e-commerce, online agricultural auctions, e-NAM, digital communication tools, agricultural e-commerce platforms, and digital farmgate systems. However, existing research also identifies several challenges, including limited digital literacy, inadequate infrastructure, fragmented platforms, logistics problems, trust issues, and lack of interoperability.

The proposed Digital Farmer Marketplace addresses this problem by providing a centralized digital platform through which farmers can list agricultural products and buyers can discover available products directly. The system aims to simplify agricultural product trading and improve farmer–buyer connectivity through a user-friendly digital marketplace.

2. PROBLEM STATEMENT

Traditional agricultural marketing systems often create difficulties for farmers in reaching suitable buyers and obtaining transparent information about product prices and demand. Farmers may depend on intermediaries for selling their produce, which can increase the distance between producers and consumers.

In addition, small and marginal farmers may have limited access to digital technologies or may find existing digital agricultural platforms difficult to use. Agricultural products also have specific requirements related to freshness, availability, transportation, and timely delivery.

Therefore, there is a need for a simple, accessible, and agriculture-specific digital marketplace that directly connects farmers and buyers while providing product information, transparent pricing, search facilities, and an organized transaction process.

3. OBJECTIVES OF THE LITERATURE SURVEY

The objectives of this literature survey are:

1. To study existing digital technologies used for agricultural marketing.
2. To analyze existing farmer-to-buyer and agricultural e-commerce platforms.
3. To identify methodologies and technologies used in previous research.
4. To study the role of e-commerce and mobile technologies in improving farmer market access.
5. To compare existing approaches based on their methodology, technology, findings, and limitations.
6. To identify challenges related to digital literacy, infrastructure, logistics, trust, and platform interoperability.
7. To identify research gaps in existing digital agricultural marketplace solutions.
8. To propose possible improvements and future research directions for a Digital Farmer Marketplace.

4. LITERATURE SEARCH METHODOLOGY

The literature survey was conducted using reputed scholarly databases and academic search platforms.

Databases Used

- Google Scholar
- ScienceDirect
- SpringerLink
- Wiley Online Library
- Frontiers
- Other academic journal repositories

Search Keywords

The following keywords were used to identify relevant research:

- "Digital Farmer Marketplace"
- "Agricultural E-commerce"
- "Farmer Buyer Digital Platform"
- "Digital Agricultural Marketplace"
- "Farmer-to-Consumer Marketplace"
- "e-NAM Farmer Market Access"
- "Digital Platforms for Farmers"
- "Agricultural Product E-commerce"
- "Digital Agriculture Marketing"
- "Smallholder Farmers Digital Market"

Research papers were selected based on their relevance to agricultural marketing, digital marketplaces, e-commerce adoption, farmer market access, and digital technologies.

5. REVIEW OF EXISTING RESEARCH PAPERS

5.1 Adoption of E-commerce Practices among Indian Farmers

Paper Title

Adoption of E-commerce Practices among the Indian Farmers: A Survey of Trichy District in the State of Tamilnadu, India

Year

2013

Research Problem

The study investigated the adoption and use of e-commerce practices among farmers in Trichy district of Tamil Nadu. The research focused on understanding whether farmers were ready to use electronic commerce for agricultural activities.

Objective

The main objective was to examine the level of e-commerce adoption among Indian farmers and understand the factors influencing their adoption of digital technologies.

Methodology

The researchers conducted a survey among farmers in the Trichy district of Tamil Nadu. The study examined farmers' awareness, usage, and adoption of e-commerce practices.

Technologies

The study focused primarily on:

- Internet technologies
- E-commerce
- Digital communication

Major Findings

The study demonstrated that e-commerce has potential to support agricultural marketing. However, technology access and farmers' willingness to adopt digital systems influence the effectiveness of such platforms.

Limitations

The research was conducted in 2013, when smartphone applications, digital payment systems, and modern cloud-based marketplaces were less developed than today. The study also focused primarily on adoption rather than developing a complete farmer-to-buyer marketplace.

Research Gap

There is a need for modern smartphone-friendly platforms that provide an integrated agricultural marketplace rather than simply providing access to e-commerce technology.

5.2 Online Agricultural Auctions in India

Paper Title

Procurement Models in the Agricultural Supply Chain: A Case Study of Online Coffee Auctions in India

Year

2007

Research Problem

Traditional agricultural procurement systems may involve multiple intermediaries and inefficient supply-chain structures. The research investigated whether online auctions could improve agricultural procurement.

Objective

The objective was to examine the use of online auctions in agricultural procurement and understand their influence on supply-chain relationships.

Methodology

The research used a case-study approach based on an online coffee auction system in India.

Technology

The primary technology studied was an **online auction/e-commerce platform**.

Major Findings

The research demonstrated that online platforms can connect agricultural producers with wider markets and improve the efficiency of procurement. However, the benefits obtained by farmers depend on the structure of the agricultural supply chain.

Limitations

The research focused specifically on coffee auctions rather than a general agricultural marketplace. It also showed that digitization alone cannot eliminate all supply-chain problems.

Research Gap

A complete agricultural marketplace should address not only online product listing but also communication, pricing, logistics, transactions, and the relationship between farmers and buyers.

5.3 Agricultural Products E-commerce Platform Based on Cloud Computing**Paper Title**

Research on the Operation of Agricultural Products E-Commerce Platform Based on Cloud Computing

Year

2022

Research Problem

Agricultural e-commerce requires efficient management of agricultural products, supply and demand information, transactions, and online services.

Objective

The study examined the operation of an agricultural products e-commerce platform and investigated the role of cloud computing in supporting digital agricultural commerce.

Methodology

The research studied the structure and operation of an agricultural e-commerce platform.

Technologies

The research focused on:

- Cloud computing
- E-commerce
- Online product search
- Supply-demand information
- Online transactions

Major Findings

Digital agricultural platforms can provide services such as product searching, supply and demand information, online shopping, seller delivery, buyer receipt, and payment.

Limitations

Agricultural products differ from ordinary commercial products because of:

- Perishability
- Freshness
- Limited shelf life
- Transportation requirements
- Seasonal availability

Research Gap

Future agricultural marketplaces should integrate agricultural logistics, product freshness, availability management, and transaction services.

5.4 Digital Technologies in Local Agri-food Systems

Paper Title

Digital Technologies in Local Agri-food Systems: Opportunities for a More Interoperable Digital Farmgate Sector

Year

2023

Research Problem

Digital agricultural platforms are increasingly being used by farmers and local food systems. However, many platforms operate independently, creating fragmentation.

Objective

The research examined the opportunities and challenges associated with digital farmgate platforms and agricultural e-commerce.

Methodology

The researchers reviewed existing literature and examined digital farmgate systems through stakeholder-focused research.

Technologies

The study considered:

- Digital platforms
- Agricultural e-commerce
- Digital farmgate systems
- Information systems

Major Findings

Digital platforms can provide significant benefits to agricultural businesses. However, fragmentation and lack of interoperability between platforms remain important problems.

Limitations

Different platforms may use different systems and data structures, making it difficult to exchange information.

Research Gap

Future agricultural platforms should support interoperability and standardized data exchange.

Relevance to Proposed Project

The proposed Digital Farmer Marketplace can be designed with modular architecture and APIs so that future payment, logistics, agricultural information, and external services can be integrated.

5.5 Digital Tools Used by Farmers in Tamil Nadu

Paper Title

Utilization Behaviour of Digital Tools by Farmers for Marketing Their Produce during Covid-19 Lockdown Period

Year

2023

Research Problem

COVID-19 restrictions created significant difficulties for farmers attempting to sell agricultural produce through traditional markets.

Objective

The study investigated how farmers used digital tools to market their agricultural products.

Study Region

The study considered farmers from:

- Krishnagiri
- Erode
- Madurai

Technologies

Farmers used:

- Telephone calls
- SMS
- WhatsApp
- Telegram

Major Findings

The study found that farmers used mobile communication technologies to contact buyers and share agricultural information during restrictions.

Limitations

General-purpose communication applications do not provide complete agricultural marketplace functionality such as structured product listings, product search, transaction management, or organized buyer discovery.

Research Gap

There is an opportunity to develop a dedicated agricultural platform that combines communication, product listing, buyer discovery, pricing, and transaction management.

5.6 Determinants of Small Farmers' Access to Agricultural Markets

Paper Title

Determinants of Small Farmers Access to Agricultural Markets: Evidence from South India

Year

2025

Research Problem

Small farmers often face difficulties accessing formal and digital agricultural markets.

Objective

The study examined the factors influencing farmers' access to agricultural markets, including awareness and adoption of digital platforms such as e-NAM.

Dataset

The study collected data from farmers, traders, and officials in South India. The research included **234 participants**, including farmers.

Methodology

The study used:

- Descriptive statistics
- ANOVA
- Binary logistic regression
-

Major Findings

Digital literacy, awareness, and knowledge of smartphones and applications influence farmers' ability to access and adopt digital agricultural platforms.

Limitations

Digital literacy, awareness, perceived risks, and regional differences can reduce adoption.

Research Gap

Agricultural marketplaces should provide a simple and accessible user interface suitable for farmers with different levels of digital literacy.

5.7 Digitalization of Agricultural Products Through E-commerce**Paper Title**

Digitalization of Agricultural Products by E-commerce Platforms in India

Year

2025

Research Problem

Indian agricultural products face difficulties related to market accessibility and supply-chain efficiency.

Objective

The research investigated how e-commerce platforms can support the digitalization and marketing of agricultural products.

Methodology

The study analyzed agricultural e-commerce and its influence on market accessibility and supply-chain processes.

Major Findings

E-commerce platforms have the potential to:

- Improve market accessibility
- Increase product visibility
- Improve supply-chain efficiency
- Provide opportunities for better farmer profitability

Limitations

Major challenges include:

- Infrastructure limitations
- Digital illiteracy
- Logistics
- Adoption barriers
- Regulatory issues

Research Gap

A successful agricultural marketplace requires both technological infrastructure and farmer-oriented usability.

5.8 Mobile Application for Real-Time Market Information and Buyer-Seller Matching

Paper Title

Mobile App for Real-Time Market Information and Buyer-Seller Matching

Year

2026

Research Problem

Farmers require real-time information about market opportunities and suitable buyers.

Objective

The proposed system aims to connect farmers and buyers through a mobile application.

Methodology

The proposed application allows farmers to:

- Register
- Upload agricultural products
- Set prices
- Display available produce

Buyers can:

- Search products
- Identify farmers
- View available commodities
- Communicate with sellers

Major Findings

A mobile platform can provide direct farmer-buyer connectivity and improve access to agricultural market information.

Limitations

The approach requires further validation in real-world agricultural environments. Challenges such as logistics, trust, digital literacy, and scalability also need to be considered.

Research Gap

An integrated platform should combine direct matching with product verification, transparent transactions, logistics, secure payments, and scalable architecture.

6. COMPARATIVE LITERATURE SURVEY

Year	Research Problem	Methodology	Key Findings	Limitations	Research Gap
2007	Agricultural procurement	Case Study	Online auctions improve market connectivity	Supply-chain dependency	Complete farmer-centric marketplace
2013	Farmer e-commerce adoption	Survey	E-commerce has agricultural potential	Older technology context	Modern mobile marketplace
2022	Agricultural e-commerce	Platform research	Supports product search and transactions	Logistics/freshness challenges	Agriculture-specific services
2023	Digital farmgate fragmentation	Literature + stakeholder research	Digital platforms provide business benefits	Poor interoperability	Standardized/interoperable platforms
2023	Farmer digital marketing	Field study	Mobile communication widely used	No dedicated marketplace	Integrated agricultural platform
2025	Small farmer market access	Survey + statistical analysis	Digital literacy affects adoption	Awareness and adoption barriers	Simple farmer-centric UI
2025	Agricultural product digitalization	Literature/secondary research	Improved accessibility and efficiency	Infrastructure /digital literacy	Inclusive marketplace
2026	Buyer-seller matching	Mobile application	Direct matching possible	Requires real-world validation	Integrated scalable marketplace

7. COMPARATIVE ANALYSIS

The reviewed research demonstrates a gradual evolution of digital agricultural marketing.

Early research primarily focused on **online auctions and e-commerce adoption**. The 2007 online coffee auction study demonstrated that online procurement can connect agricultural producers with wider markets. The 2013 Trichy study further highlighted the importance of farmer adoption of e-commerce.

Later research shifted toward **integrated agricultural e-commerce platforms**. Cloud-based agricultural platforms introduced online product search, supply-demand information, purchasing, payment, and delivery-related functionality.

More recent research has focused on **digital farmgate systems, e-NAM adoption, digital literacy, and direct buyer-seller matching**. These studies show that technology alone is insufficient. Successful adoption also depends on usability, awareness, infrastructure, trust, and interoperability.

Therefore, existing approaches can be broadly classified into:

1. Online agricultural auctions
2. General agricultural e-commerce
3. Government digital market platforms
4. Mobile agricultural information systems
5. Digital farmgate platforms
6. Direct farmer-buyer marketplace applications

The proposed Digital Farmer Marketplace builds upon these approaches by focusing specifically on **direct farmer-buyer connectivity and a simple user-oriented marketplace interface**.

8 LIMITATIONS IDENTIFIED FROM EXISTING RESEARCH

The literature review identifies the following major limitations.

8.1 Digital Literacy

Many farmers may not have sufficient knowledge of smartphones, applications, or online transactions.

8.2 Infrastructure

Poor internet connectivity and limited digital infrastructure can restrict access to digital agricultural platforms.

8.3 Platform Fragmentation

Different agricultural platforms may operate independently and may not exchange information effectively.

8.4 Logistics

Agricultural products are often perishable and require timely transportation.

8.5 Trust and Verification

Online buyers need confidence regarding:

- Farmer identity
- Product quality
- Product quantity

- Price
- Transaction reliability

8.6 Limited Direct Interaction

Some systems focus primarily on market information or auctions rather than establishing a complete farmer-buyer relationship.

8.7 Adoption Barriers

Even technically capable platforms may experience low adoption because of:

- Lack of awareness
- Complexity
- Perceived risk
- Lack of training

8.8 Lack of Personalization

Existing systems may provide general market information without personalized recommendations based on location, product availability, demand, or buyer preferences.

9. RESEARCH GAP

Based on the literature survey, several research gaps have been identified.

Although digital agricultural platforms have demonstrated the ability to improve market access, existing approaches continue to face challenges related to digital literacy, platform fragmentation, logistics, trust, interoperability, and usability.

Many existing systems concentrate on individual aspects such as:

- market information,
- online auctions,
- product listing,
- e-commerce,
- government market access,
- or communication.

However, there is still scope for an integrated and farmer-friendly platform that combines these functions into a single system.

The major research gap identified is therefore:

There is a need for a simple, scalable, agriculture-specific digital marketplace that provides direct farmer-buyer connectivity while integrating product listing, transparent pricing, product discovery,

communication, transaction support, and future logistics and payment services, without creating excessive complexity for farmers with limited digital literacy.

This gap forms the foundation of the proposed Digital Farmer Marketplace.

10. PROPOSED SYSTEM BASED ON THE RESEARCH GAP

The proposed Digital Farmer Marketplace aims to address the identified research gaps through a centralized web-based platform.

Farmer Side

Farmers can:

- Create an account
- Create a farmer profile
- Add agricultural products
- Specify price
- Specify quantity
- Update product availability
- View buyer interactions

Buyer Side

Buyers can:

- Register/login
- Browse agricultural products
- Search products
- Filter products
- View product information
- View farmer information
- Contact farmers
- Place orders in future versions

Marketplace Layer

The platform provides:

- Product catalogue
- Search
- Product categories
- Price information
- Availability information
- Farmer profiles

Future Integration

The system can later integrate:

- Online payments
- GPS/location services
- Delivery tracking
- Logistics providers
- AI price prediction
- Demand forecasting
- Recommendation systems
- Product traceability

11. FUTURE RESEARCH DIRECTIONS

11.1 AI-Based Price Prediction

Machine-learning models can analyze historical market prices and recommend suitable prices to farmers.

11.2 Demand Forecasting

Historical transactions, seasonal trends, weather information, and market demand can be analyzed to predict future demand.

11.3 Recommendation System

The system could recommend:

- products to buyers,
- buyers to farmers,
- products based on location,
- products based on previous purchases.

11.4 Multilingual Interface

Supporting regional languages can improve accessibility for farmers.

For example:

- Tamil
- Telugu
- Hindi
- Kannada
- Malayalam

11.5 Voice-Based Interaction

Voice interfaces could help farmers with limited literacy or difficulty typing.

11.6 Secure Digital Payments

Future versions can integrate secure digital payment systems.

11.7 Logistics Integration

The platform can connect farmers with transportation providers and support:

- pickup scheduling,
- delivery tracking,
- location sharing.

11.8 Product Traceability

Blockchain or other distributed-ledger technologies could potentially be used to maintain transparent product histories.

11.9 IoT-Based Freshness Monitoring

IoT sensors could be integrated into future supply chains to monitor:

- temperature,
- humidity,
- storage conditions,
- transportation conditions.

12. CONCLUSION

The literature survey demonstrates that digital technologies have significant potential to transform agricultural marketing by improving market access, information availability, communication, and farmer-buyer connectivity.

Previous studies have explored online agricultural auctions, e-commerce adoption, cloud-based agricultural platforms, digital farmgate systems, e-NAM, mobile communication tools, and direct buyer-seller matching. These studies demonstrate the benefits of digital agricultural systems but also identify significant challenges related to digital literacy, infrastructure, logistics, trust, adoption, and platform interoperability.

The review indicates that a successful digital agricultural marketplace must go beyond simple product listing. It should provide a **simple, accessible, transparent, and integrated environment** for farmers and buyers.

The proposed **Digital Farmer Marketplace** addresses this identified research gap by focusing on direct farmer-buyer connectivity and providing a centralized platform for agricultural product discovery and management. Future extensions involving AI-based price prediction, demand forecasting, multilingual interfaces, digital payments, logistics integration, and product traceability can further improve the platform.

Thus, the literature supports the relevance of developing a farmer-oriented digital marketplace while also providing clear directions for future research and technological enhancement.

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